A Computations

Swath width [17]:

$$W = 2h \tan\left(\frac{\alpha_{\text{FOV}}}{2}\right)$$

Right Ascension of the Ascending node - This is the angular separation between the orbital planes of the 20 CubeSats:

$$\text{RAAN spacing} = \frac{360^\circ}{20} = 18^\circ = \Delta\lambda = 0.314 \text{ rad}$$

Assuming $\alpha_{\text{FOV}} = 3.667^\circ$, we can calculate different values of W for different values of h:

Table 3: Table of values of h and W

$h = 100$ km	$W = 6.4$ km
$h = 200$ km	$W = 12.8$ km
$h = 250$ km	$W = 16.0$ km

The number of strips needed to be covered for full global coverage at varying altitudes is therefore the moon's surface area divided by the surface area of 1 strip:

$$N_{\text{strips}} = \frac{A_{\text{moon}}}{A_{1\text{ strip}}} = \frac{4\pi R_{\text{M}}^2}{W \times (2\pi R_{\text{M}})} = \frac{2R_{\text{M}}}{W} = \frac{2 \times 1737}{W} = \frac{3474}{W}$$

Table 4: Number of strips for different altitudes

h (km)	N_{strips}
100	542.8
200	271.4
250	217.13

The time taken for a CubeSat to make one full orbital revolution (Orbit Period) at different orbital radii, where Orbit Radius (r) is the moon's radius plus the CubeSat's altitude, and where μ_M is the Moon's Gravitational Constant [17]:

$$T = 2\pi\sqrt{\frac{r^3}{\mu_M}} \quad \text{with} \quad r = R_M + h, \quad \mu_M = 4902.8 \text{ km}^3/\text{s}^2$$

Table 5: Orbital radii and corresponding periods

Orbit	r (km)	T (min)
1	1837	117.75
2	1937	127.50
3	1987	132.47

The number of strips covered per day by the constellation due to the Orbital Period at different altitudes is therefore:

$$\text{Strips per day} \approx N_{\text{sats}} \times \frac{1440}{T_{\text{min}}}$$

The total time taken for full global coverage is therefore the number of strips needed to be covered

divided by the actual number of strips covered per day:

$$t_{\text{cov}} = \frac{\text{number of strips}}{\text{strips per day}}$$

Table 6: Time for full coverage for different altitudes

h (km)	t_{cov} (days)
100	2.21
200	1.20
250	0.99

Since this is a feasibility study, it is best practice to keep in mind that these orbits will be perturbed by lunar mascons. This is taken into account by the realism factor k:

$$k \sim 1.2 - 1.5$$

$$t_{\text{real}} \approx k\, t_{\text{cov}}$$

Table 7: Final realistic global time coverage for different altitudes

Altitude h (km)	$t_{\text{real,min}}$(days)	$t_{\text{real,max}}$(days)
100	2.65	3.32
200	1.44	1.80
250	1.19	1.49

From the table, 250 km exhibits the closest approximation to daily global revisit, making it the best candidate for our imaging layer of 20 CubeSats.